



# PSOC™ Edge Protect Bootloader

## Customer training workshop

December 2025



# Prerequisites

## Training prerequisites:

- Introduction to PSOC™ Edge and ModusToolbox™
  - For an introduction to PSOC™ MCUs, including a getting started guide to ModusToolbox™, visit:  
<https://www.infineon.com/product-information/psocdeveloper>
  - For PSOC™ Edge trainings, from beginner tutorials to advanced trainings, visit: [PSOC™ Edge E84 Training Collection](#)

## – Development tools:

- ModusToolbox™ v3.6
- Eclipse IDE for ModusToolbox™ v2025.4.0
- Edge Protect Security Suite v1.6
- ModusToolbox™ Programming Tools v1.5.0
- KIT\_PSE84\_EVAL\_EPC2 BSP v1.0.0
- Terminal Emulator
- KIT\_PSE84\_EVAL

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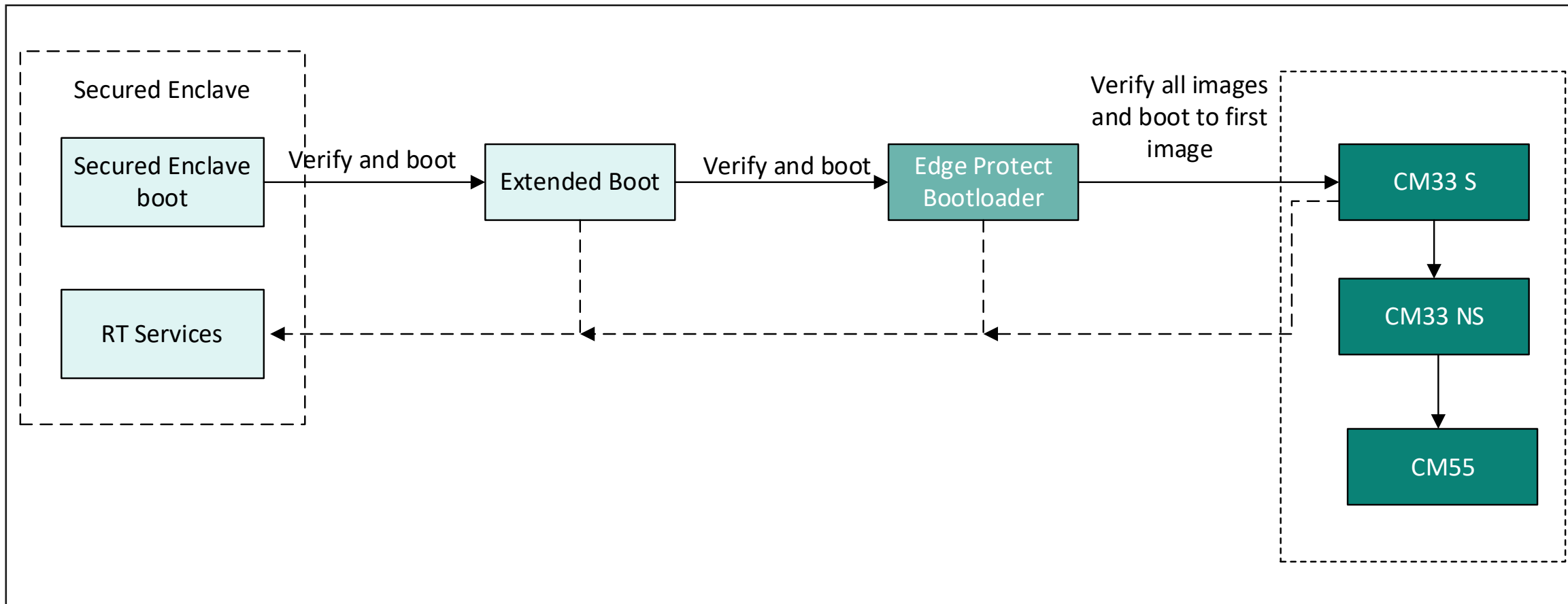
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# Introduction to Edge Protect Bootloader (EPB)

- Edge Protect Bootloader (EPB) is an open-source [MCUboot](#) library-based bootloader for PSOC™ Edge MCU
- This is an optional bootloader stage, owned by the OEMs. This is the first OEM code that is executed, when present in the boot chain
- Infineon provides the EPB as a code example along with the open-source MCUboot library
- EPB extends the chain of trust (CoT) from the extended boot to the rest of the applications in the system, i.e., extended boot verifies and loads EPB; and the EPB verifies and loads the rest of the images
- In default configurations, the EPB can verify up to three images and launch the next image in the boot sequence
- EPB is designed to run in the secure mode of the CPU, with all resource access privileges available to it. It is hardened against physical attacks using the FIH library. This is the same fault injection hardening (FIH) library that is used by open-source MCUboot

# Chain of trust (CoT)

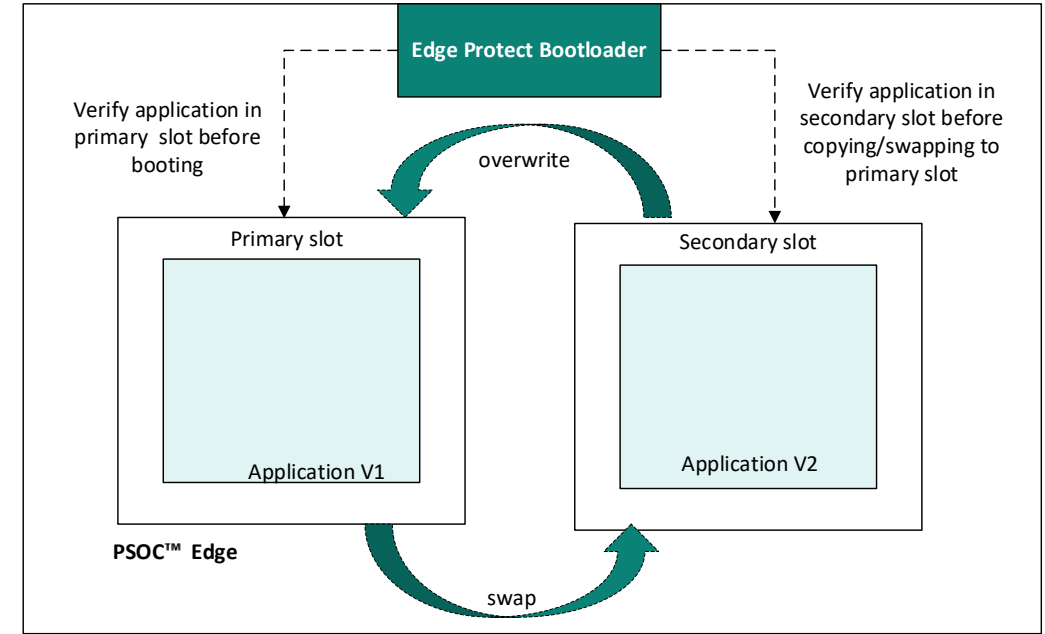


# Secure boot of Edge Protect Bootloader

- For a complete CoT, it is essential that Edge Protect Bootloader is authenticated by the extended boot, before launching
- Before launching, the extended boot authenticates the Edge Protect Bootloader using ECDSA P256. It uses OEM\_ROT\_KEY (public) provisioned to the device for image authentication
- Edge Protect Bootloader must be signed using OEM\_ROT\_KEY (private)

# Memory map

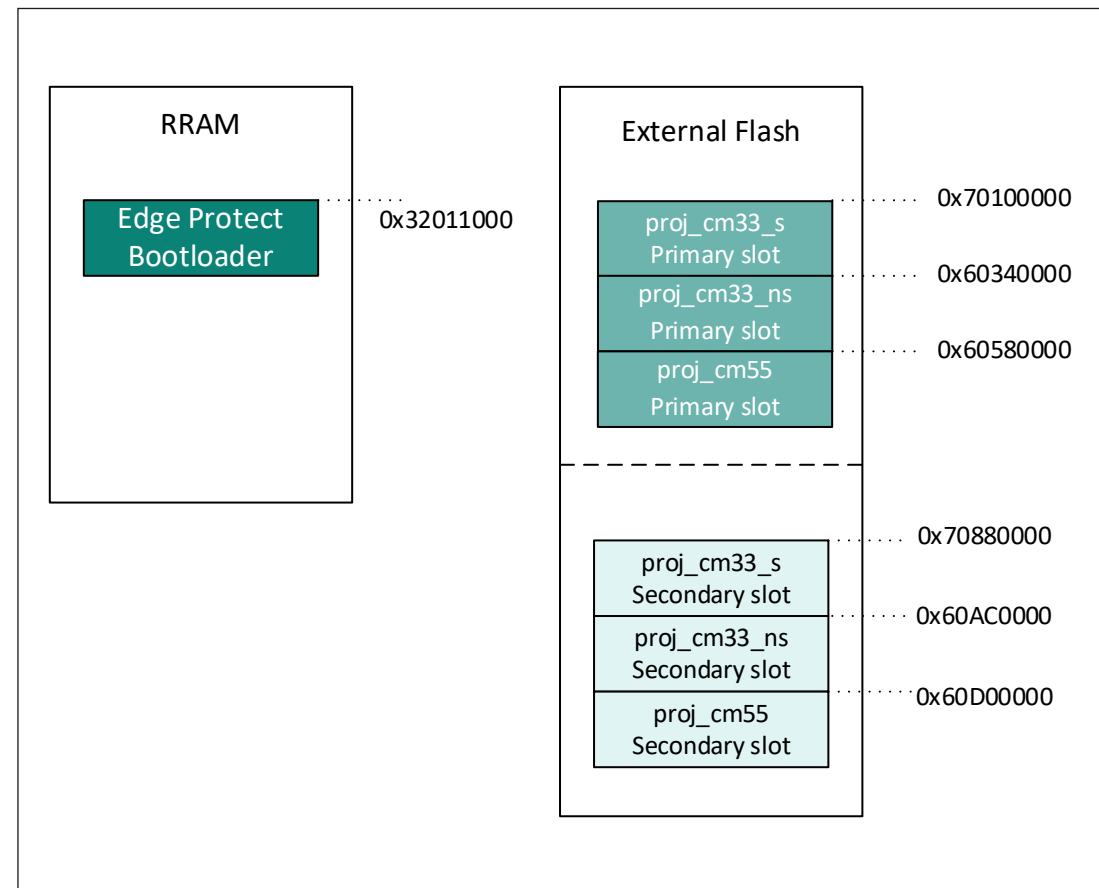
- The flash memory can be partitioned into multiple image areas; each containing two image slots:
  - primary slot
  - secondary slot
- EPB validates and executes application from the primary slot
- EPB validates upgrade image in secondary slot and copies it to primary slot for execution





## Memory map (contd.)

- Memory map describes allocated memory areas for different application slots and defines their start offsets and sizes
- It is defined at compile time and cannot be changed dynamically
- When the flash memory contains multiple executable images , the flash must be partitioned to allocate two slots for each image



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# Image validation

- During the normal boot, EPB validates that the image is in [MCUboot image format](#)
- When secure boot is enabled in the EPB, it performs additional signature verification of the images
- EPB uses OEM\_ROT\_KEY provisioned in the device to verify the application images
- EPB makes use of the basic runtime (RT) services running on Secure Enclave to authenticate the application using ECDSA P256

## Image validation (contd.)

- Secure boot in EPB can be enabled by enabling the "Validate boot slot" and "Validate upgrade slot" options in the bootloader solution. Provide the path to OEM private key to be used to sign the application.

Edge Protect Bootloader Solution - Parameters

Enter filter text...

| Name                        | Value                                                |
|-----------------------------|------------------------------------------------------|
| > Documentation             |                                                      |
| ▼ MCUBoot config            |                                                      |
| (?) Upgrade mode            | overwrite                                            |
| (?) Logging level           | dbg                                                  |
| (?) FIH level               | off                                                  |
| (?) Image count             | 3                                                    |
| (?) Validate boot slot      | <input checked="" type="checkbox"/>                  |
| (?) Validate upgrade slot   | <input checked="" type="checkbox"/>                  |
| (?) Measured boot           | <input type="checkbox"/>                             |
| (?) HW rollback protection  | <input type="checkbox"/>                             |
| (?) Dependency check        | <input type="checkbox"/>                             |
| (?) SW downgrade prevention | <input type="checkbox"/>                             |
| > Encryption                |                                                      |
| (?) Stage SE RAM App        | <input type="checkbox"/>                             |
| (?) Watchdog                | <input type="checkbox"/>                             |
| ▼ Bootloader Configuration  |                                                      |
| (?) Bootloader project name | proj_bootloader                                      |
| (?) Input hex file path     | ../..../build/project_hex/proj_bootloader.hex        |
| (?) Output hex file path    | ../..../build/project_hex/proj_bootloader_signed.hex |
| (?) OEM private key path    | ../..../keys/oem_private_key_0.pem                   |
| (?) Bootloader NVM Region   | <input checked="" type="checkbox"/>                  |

## Image upgrade

- For firmware update use case, upgrade images are downloaded to secondary slots by user application
- EPB detects the new image in the upgrade slot, validates it, copies to the primary slot and launches the application
- EPB supports two different image upgrade methods:
  - Upgrade by overwrite
  - Upgrade by swap
- Either of these methods can be configured at compile time and the compiled bootloader will only support the selected upgrade method
- The update process is power fail safe. If power failure happens in the middle of an update, Edge Protect Bootloader will perform the update on next reset of the device

# Image upgrade (contd.)

- To enable the image upgrade using MTB, create the upgrade slots for each image in the Memory tab of the device configurator as shown

The screenshot shows the Memory tab of the device configurator. It displays various memory regions including ITCM, DTCM, RRAM, SRAM, System SRAM (SoCMEM), and Serial Memory Interface block 0, memory 1 (S25FS128S). Below the regions, a table lists memory regions with columns for Region Id, Domain, Offset, Size, and Description. The 'm33s\_secondary', 'm33s\_secondary', and 'm55\_secondary' regions are highlighted with a red box. Below this table, another table titled 'Memory Ranges' shows the definition of primary slots for project 0, project 1, and project 2.

| Region Id      | Domain             | Offset     | Size                          | Description                                    |
|----------------|--------------------|------------|-------------------------------|------------------------------------------------|
| UNALLOCATED    | <NONE>             | 0x00000000 | 0x00100000 (1 MB)             |                                                |
| m33s_nvmm      | M33S (Domain 0)    | 0x00100000 | 0x00200000 (2 MB)             | CM33 secure image including non-secure callabl |
| m33s_trailer   | M33S (Domain 0)    | 0x00300000 | 0x00040000 (256 KB)           | CM33 secure trailer                            |
| m33s_nvmm      | M33 (Domain 1)     | 0x00340000 | 0x00200000 (2 MB)             | CM33 image                                     |
| m33s_trailer   | M33 (Domain 1)     | 0x00540000 | 0x00040000 (256 KB)           | CM33 trailer                                   |
| m55_nvmm       | M55 (Domain 2)     | 0x00580000 | 0x002C0000 (2.75 MB)          | CM55 image                                     |
| m55_trailer    | M55 (Domain 2)     | 0x00840000 | 0x00040000 (256 KB)           | CM55 trailer                                   |
| m33s_secondary | M33S (Domain 0)    | 0x00880000 | 0x00240000 (2.25 MB)          |                                                |
| m33s_secondary | M33 (Domain 1)     | 0x00AC0000 | 0x00240000 (2.25 MB)          |                                                |
| m55_secondary  | M33_M55 (Domain 4) | 0x00D00000 | 0x00300000 (3 MB)             |                                                |
|                |                    | 0x01000000 | USED: (15 MB)<br>FREE: (1 MB) |                                                |

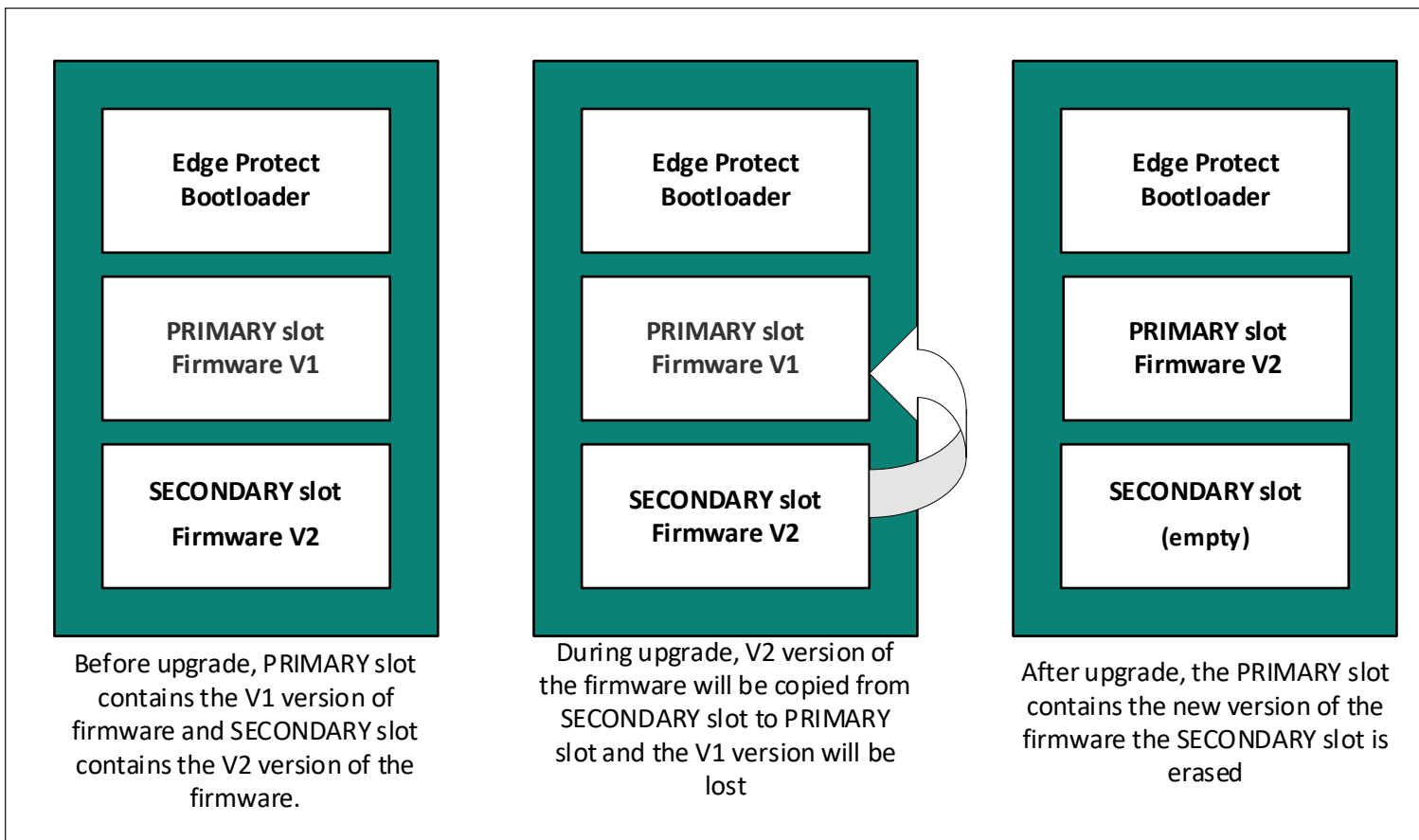
  

| Range Id           | Offset     | Size                 | Included Regions | Description                             |
|--------------------|------------|----------------------|------------------|-----------------------------------------|
| app_0_primary_slot | 0x00100000 | 0x00200000 (2 MB)    | m33s_nvmm        | Defintion of primary slot for project 0 |
| app_1_primary_slot | 0x00340000 | 0x00200000 (2 MB)    | m33s_nvmm        | Defintion of primary slot for project 1 |
| app_2_primary_slot | 0x00580000 | 0x002C0000 (2.75 MB) | m55_nvmm         | Defintion of primary slot for project 2 |

The screenshot shows the 'Edge Protect Bootloader Solution - Parameters' dialog. It contains configuration parameters for three projects. The 'Secondary slot region' is set to 'm33s\_upgrade\_slot (Memory Region 23)' for Project 0, 'm33s\_upgrade\_slot (Memory Region 24)' for Project 1, and 'm55\_upgrade\_slot (Memory Region 25)' for Project 2. The 'Primary slot memory' is set to 'Serial Memory Interface block 0, memory 1 (S25FS128S)' for all three projects. The 'Primary slot offset' and 'Primary slot size' are also configured for each project. The 'Secure Access' checkbox is checked for Project 0 and Project 1, and unchecked for Project 2.

| Name                    | Value                                                 |
|-------------------------|-------------------------------------------------------|
| Project 0 Configuration |                                                       |
| Secondary slot region   | m33s_upgrade_slot (Memory Region 23)                  |
| Primary slot memory     | Serial Memory Interface block 0, memory 1 (S25FS128S) |
| Primary slot offset     | 0x00140000                                            |
| Primary slot size       | 0x00200000                                            |
| Secure Access           | <input checked="" type="checkbox"/>                   |
| Project 1 Configuration |                                                       |
| Secondary slot region   | m33s_upgrade_slot (Memory Region 24)                  |
| Primary slot memory     | Serial Memory Interface block 0, memory 1 (S25FS128S) |
| Primary slot offset     | 0x00340000                                            |
| Primary slot size       | 0x00240000                                            |
| Secure Access           | <input type="checkbox"/>                              |
| Project 2 Configuration |                                                       |
| Secondary slot region   | m55_upgrade_slot (Memory Region 25)                   |
| Primary slot memory     | Serial Memory Interface block 0, memory 1 (S25FS128S) |
| Primary slot offset     | 0x00580000                                            |
| Primary slot size       | 0x00300000                                            |
| Secure Access           | <input type="checkbox"/>                              |

# Upgrade by overwrite



## Upgrade by overwrite (contd.)

- This upgrade method can be selected in the bootloader solution personality
- The secondary slots for each image must also be allocated in the Memory tab of the device configurator and linked in the bootloader solution personality
- In your MTB application's common.mk file , use the bootloader solution generated JSON file for boot and upgrade images

◦ For **BOOT\_IMAGE** : Combine application projects and EdgeProtect bootloader project for initial programming and booting.

```
COMBINE_SIGN_JSON?=./bsps/TARGET_${TARGET}/config/GeneratedSource/boot_with_blldr.json
```

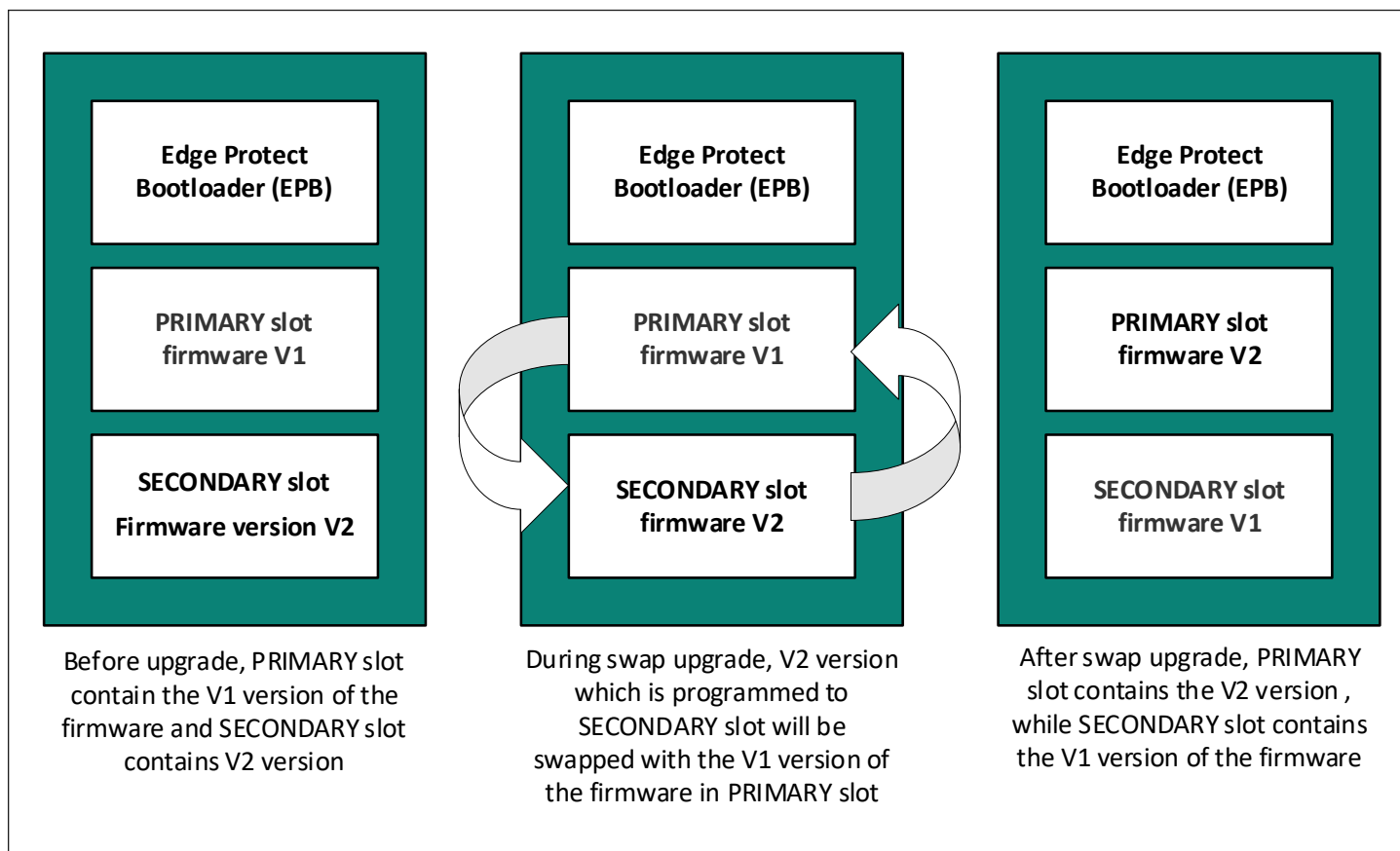
◦ For **UPDATE\_IMAGE** : Combine only the application projects for update

```
COMBINE_SIGN_JSON?=./bsps/TARGET_${TARGET}/config/GeneratedSource/boot_with_blldr_upgr.json
```

| Edge Protect Bootloader Solution - Parameters |                          |                                   |
|-----------------------------------------------|--------------------------|-----------------------------------|
| Enter filter text...                          |                          |                                   |
| Name                                          | Value                    |                                   |
| > Documentation                               |                          |                                   |
| ▼ MCUBoot config                              |                          |                                   |
| ? Upgrade mode                                | overwrite                |                                   |
| ? Logging level                               | dbg                      |                                   |
| ? FIH level                                   | off                      |                                   |
| ? Image count                                 | 3                        |                                   |
| ? Validate boot slot                          | <input type="checkbox"/> |                                   |
| ? Validate upgrade slot                       | <input type="checkbox"/> |                                   |
| ? Measured boot                               | <input type="checkbox"/> |                                   |
| ? HW rollback protection                      | <input type="checkbox"/> |                                   |
| ? Dependency check                            | <input type="checkbox"/> |                                   |
| ? SW downgrade prevention                     | <input type="checkbox"/> |                                   |
| ? Stage SE RAM App                            | <input type="checkbox"/> |                                   |
| ? Watchdog                                    | <input type="checkbox"/> |                                   |
| > Bootloader Configuration                    |                          |                                   |
| ▼ Project 0 Configuration                     |                          |                                   |
| ▼ Memory Slots                                |                          |                                   |
| ? Secondary slot region                       | <input type="checkbox"/> | m33s_secondary (Memory Region 25) |
| > RAM Load                                    |                          |                                   |
| > Build & Deployment                          |                          |                                   |
| > Primary Slot Configuration                  |                          |                                   |
| ▼ Project 1 Configuration                     |                          |                                   |
| ▼ Memory Slots                                |                          |                                   |
| ? Secondary slot region                       | <input type="checkbox"/> | m33_secondary (Memory Region 26)  |
| > RAM Load                                    |                          |                                   |
| > Build & Deployment                          |                          |                                   |
| > Primary Slot Configuration                  |                          |                                   |
| ▼ Project 2 Configuration                     |                          |                                   |
| ▼ Memory Slots                                |                          |                                   |
| ? Secondary slot region                       | <input type="checkbox"/> | m55_secondary (Memory Region 27)  |
| > RAM Load                                    |                          |                                   |
| > Build & Deployment                          |                          |                                   |
| > Primary Slot Configuration                  |                          |                                   |



# Upgrade by swap



## Upgrade by swap (contd.)

- To enable swap upgrade in MTB application, create new memory regions for swap\_status and swap\_scratch region as shown below:

RRAM

|  | Region Id              | Domain              | Offset     | Size                             | Description |
|--|------------------------|---------------------|------------|----------------------------------|-------------|
|  | extended_boot_reserved | reserved (Domain 5) | 0x00000000 | 0x00011000 (68 KB)               |             |
|  | bootloader_nvm         | M33S (Domain 0)     | 0x00011000 | 0x00028000 (160 KB)              |             |
|  | swap_status            | M33S (Domain 0)     | 0x00039000 | 0x00004000 (16 KB)               |             |
|  | UNALLOCATED            | <NONE>              | 0x0003D000 | 0x0002D000 (180 KB)              |             |
|  | reserved_region        | reserved (Domain 5) | 0x0006A000 | 0x00016000 (88 KB)               |             |
|  |                        |                     | 0x00080000 | USED: (332 KB)<br>FREE: (180 KB) |             |

SRAM

System SRAM (SoCMEM)

Serial Memory Interface block 0, memory 1 (S25FS128S)

|  | Region Id         | Domain          | Offset     | Size                               | Description                                            |
|--|-------------------|-----------------|------------|------------------------------------|--------------------------------------------------------|
|  | swap_scratch      | M33S (Domain 0) | 0x00000000 | 0x00040000 (256 KB)                |                                                        |
|  | UNALLOCATED       | <NONE>          | 0x00040000 | 0x000C0000 (768 KB)                |                                                        |
|  | m33s_nvm          | M33S (Domain 0) | 0x00100000 | 0x00200000 (2 MB)                  | CM33 secure image including non-secure callable region |
|  | m33s_trailer      | M33S (Domain 0) | 0x00300000 | 0x00040000 (256 KB)                | CM33 secure trailer                                    |
|  | m33_nvm           | M33 (Domain 1)  | 0x00340000 | 0x00200000 (2 MB)                  | CM33 image                                             |
|  | m33_trailer       | M33 (Domain 1)  | 0x00540000 | 0x00040000 (256 KB)                | CM33 trailer                                           |
|  | m55_nvm           | M55 (Domain 2)  | 0x00580000 | 0x002C0000 (2.75 MB)               | CM55 image                                             |
|  | m55_trailer       | M55 (Domain 2)  | 0x00840000 | 0x00040000 (256 KB)                | CM55 trailer                                           |
|  | m33s_upgrade_slot | M33S (Domain 0) | 0x00880000 | 0x00240000 (2.25 MB)               | secondary slot for cm33 secure app                     |
|  | cm33_upgrade_slot | M33 (Domain 1)  | 0x00AC0000 | 0x00240000 (2.25 MB)               | secondary slot for CM33 non secure app                 |
|  | cm55_upgrade_slot | M55 (Domain 2)  | 0x00D00000 | 0x00300000 (3 MB)                  | secondary slot for CM55 app                            |
|  |                   |                 | 0x01000000 | USED: (15.25 MB)<br>FREE: (768 KB) |                                                        |

## Upgrade by swap (contd.)

- This upgrade method can be selected in the bootloader solution personality
- The secondary slots for each image, swap status and swap scratch regions must be linked in the bootloader solution personality
- In your MTB application's common.mk file , use the bootloader solution generated JSON file for boot and upgrade images

- For **BOOT\_IMAGE** : Combine application projects and EdgeProtect bootloder project for initial programming and booting.

```
COMBINE_SIGN_JSON?=./bsps/TARGET_${TARGET}/config/GeneratedSource/boot_with_bldr.json
```

- For **UPDATE\_IMAGE** : Combine only the application projects for update

```
COMBINE_SIGN_JSON?=./bsps/TARGET_${TARGET}/config/GeneratedSource/boot_with_bldr_upgr.json
```

| Name                                  | Value                                                         |
|---------------------------------------|---------------------------------------------------------------|
| Documentation                         |                                                               |
| Edge Protect Bootloader documentation | <a href="#">EdgeProtect Bootloader Configuration Overview</a> |
| MCUBoot config                        |                                                               |
| Upgrade mode                          | swap                                                          |
| Logging level                         | dbg                                                           |
| FIH level                             | off                                                           |
| Image count                           | 3                                                             |
| Validate boot slot                    | <input type="checkbox"/>                                      |
| Validate upgrade slot                 | <input type="checkbox"/>                                      |
| Measured boot                         | <input type="checkbox"/>                                      |
| HW rollback protection                | <input type="checkbox"/>                                      |
| Dependency check                      | <input type="checkbox"/>                                      |
| Stage SE RAM App                      | <input type="checkbox"/>                                      |
| Watchdog                              | <input type="checkbox"/>                                      |
| Bootloader Application Configuration  |                                                               |
| Bootloader project name               | proj_bootloader                                               |
| Input hex file path                   | ../././././build/project_hex/proj_bootloader.hex              |
| Output hex file path                  | ../././././build/project_hex/proj_bootloader_signed.hex       |
| OEM private key path                  | ../././././keys/oem_private_key_0.pem                         |
| Bootloader NVM Region                 | <input checked="" type="checkbox"/>                           |
| Bootloader SRAM load                  | <input type="checkbox"/>                                      |
| Enable Bootloader Signing             | <input type="checkbox"/>                                      |
| Swap Status Region                    | <input type="checkbox"/> swap_status (Memory Region 28)       |
| Scratch Region                        | <input type="checkbox"/> swap_scratch (Memory Region 27)      |

# Downgrade prevention

- Downgrade prevention is a feature which ensures that only firmware with a higher version or security counter can replace an existing image
- This safeguards against malicious attempts to revert the device to a potentially vulnerable older version
- There are two types of downgrade prevention
  - Software-based downgrade prevention
  - Hardware-based downgrade prevention

# Software-based downgrade prevention

- Software-based downgrade prevention compares image version numbers to block installations of older firmware.
- This feature works exclusively with the overwrite-based image update strategy.
- EPB will reject the image upgrade if the new image contains lower major/ minor/patch version

| Current image version | Upgrade image version | EPB action       |
|-----------------------|-----------------------|------------------|
| 1.2.3                 | 1.2.2                 | Upgrade rejected |

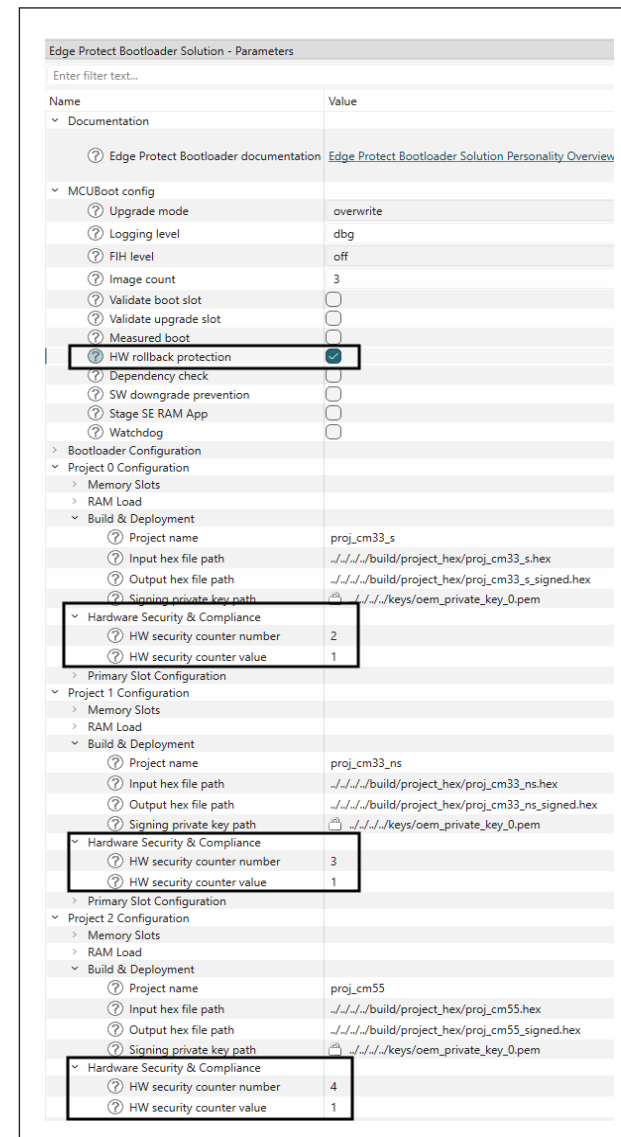
Edge Protect Bootloader Solution - Parameters

Enter filter text...

| Name                                  | Value                                                                 |
|---------------------------------------|-----------------------------------------------------------------------|
| Documentation                         |                                                                       |
| Edge Protect Bootloader documentation | <a href="#">Edge Protect Bootloader Solution Personality Overview</a> |
| MCUBoot config                        |                                                                       |
| Upgrade mode                          | overwrite                                                             |
| Logging level                         | dbg                                                                   |
| FIH level                             | off                                                                   |
| Image count                           | 3                                                                     |
| Validate boot slot                    | <input type="checkbox"/>                                              |
| Validate upgrade slot                 | <input type="checkbox"/>                                              |
| Measured boot                         | <input type="checkbox"/>                                              |
| HW rollback protection                | <input type="checkbox"/>                                              |
| Dependency check                      | <input type="checkbox"/>                                              |
| SW downgrade prevention               | <input checked="" type="checkbox"/>                                   |
| Stage SE RAM App                      | <input type="checkbox"/>                                              |
| Watchdog                              | <input type="checkbox"/>                                              |
| Bootloader Configuration              |                                                                       |
| Project 0 Configuration               |                                                                       |
| Memory Slots                          |                                                                       |
| RAM Load                              |                                                                       |
| Build & Deployment                    |                                                                       |
| Project name                          | proj_cm33_s                                                           |
| Input hex file path                   | ../build/project_hex/proj_cm33_s.hex                                  |
| Output hex file path                  | ../build/project_hex/proj_cm33_s_signed.hex                           |
| Signing private key path              | ../keys/oem_private_key_0.pem                                         |
| Image version                         | 1.0.0                                                                 |
| Primary Slot Configuration            |                                                                       |
| Project 1 Configuration               |                                                                       |
| Memory Slots                          |                                                                       |
| RAM Load                              |                                                                       |
| Build & Deployment                    |                                                                       |
| Project name                          | proj_cm33_ns                                                          |
| Input hex file path                   | ../build/project_hex/proj_cm33_ns.hex                                 |
| Output hex file path                  | ../build/project_hex/proj_cm33_ns_signed.hex                          |
| Signing private key path              | ../keys/oem_private_key_0.pem                                         |
| Image version                         | 1.0.0                                                                 |
| Primary Slot Configuration            |                                                                       |
| Project 2 Configuration               |                                                                       |
| Memory Slots                          |                                                                       |
| RAM Load                              |                                                                       |
| Build & Deployment                    |                                                                       |
| Project name                          | proj_cm55                                                             |
| Input hex file path                   | ../build/project_hex/proj_cm55.hex                                    |
| Output hex file path                  | ../build/project_hex/proj_cm55_signed.hex                             |
| Signing private key path              | ../keys/oem_private_key_0.pem                                         |
| Image version                         | 1.0.0                                                                 |
| Primary Slot Configuration            |                                                                       |

# Hardware-based downgrade prevention

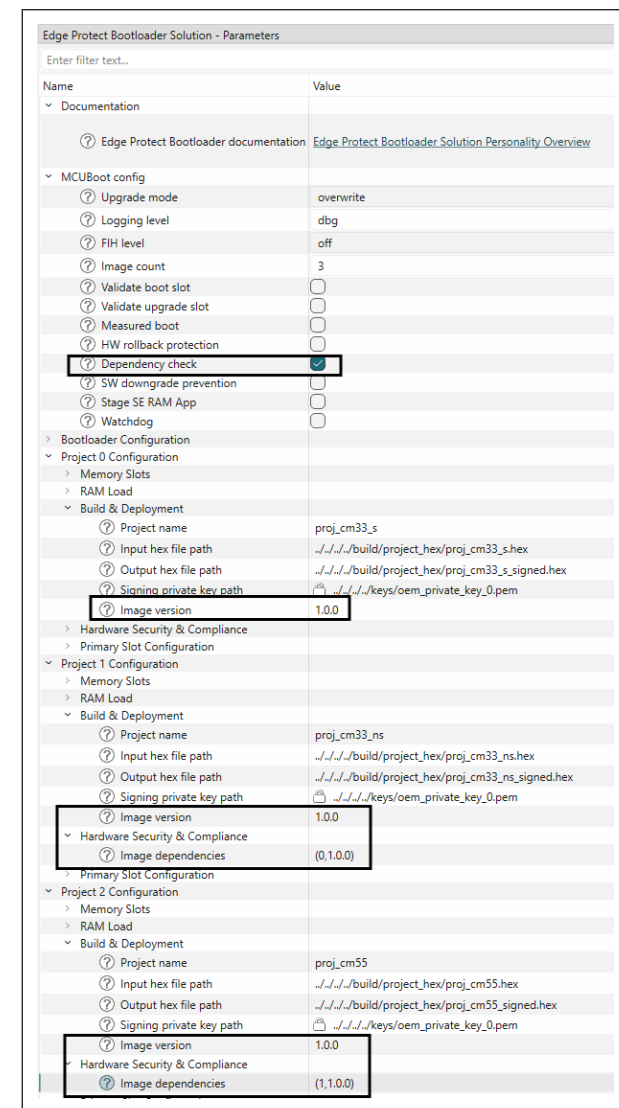
- Each signed image can contain a security counter in its protected TLV area
- During the hardware-based downgrade prevention (rollback protection) the new image's security counter value will be compared with the currently active security counter (stored in the Secure Enclave)
- “HW security counter” number refers to the number of the counter cell in the device. It can be set in the range of 2-7
- “HW security counter value” refers to the counter value that is stored in the counter. It can be set in the range of 1-63 or left unchanged



| Name                                  | Value                                                                 |
|---------------------------------------|-----------------------------------------------------------------------|
| Documentation                         |                                                                       |
| Edge Protect Bootloader documentation | <a href="#">Edge Protect Bootloader Solution Personality Overview</a> |
| MCUBoot config                        |                                                                       |
| Upgrade mode                          | overwrite                                                             |
| Logging level                         | dbg                                                                   |
| FIH level                             | off                                                                   |
| Image count                           | 3                                                                     |
| Validate boot slot                    | <input type="checkbox"/>                                              |
| Validate upgrade slot                 | <input type="checkbox"/>                                              |
| Measured boot                         | <input type="checkbox"/>                                              |
| HW rollback protection                | <input checked="" type="checkbox"/>                                   |
| Dependency check                      | <input type="checkbox"/>                                              |
| SW downgrade prevention               | <input type="checkbox"/>                                              |
| Stage SE RAM App                      | <input type="checkbox"/>                                              |
| Watchdog                              | <input type="checkbox"/>                                              |
| Bootloader Configuration              |                                                                       |
| Project 0 Configuration               |                                                                       |
| Memory Slots                          |                                                                       |
| RAM Load                              |                                                                       |
| Build & Deployment                    |                                                                       |
| Project name                          | proj_cm33_s                                                           |
| Input hex file path                   | ./.././././build/project_hex/proj_cm33_s.hex                          |
| Output hex file path                  | ./.././././build/project_hex/proj_cm33_s_signed.hex                   |
| Signing private key path              | ./.././././keys/oem_private_key_0.pem                                 |
| Hardware Security & Compliance        |                                                                       |
| HW security counter number            | 2                                                                     |
| HW security counter value             | 1                                                                     |
| Project 1 Configuration               |                                                                       |
| Memory Slots                          |                                                                       |
| RAM Load                              |                                                                       |
| Build & Deployment                    |                                                                       |
| Project name                          | proj_cm33_ns                                                          |
| Input hex file path                   | ./.././././build/project_hex/proj_cm33_ns.hex                         |
| Output hex file path                  | ./.././././build/project_hex/proj_cm33_ns_signed.hex                  |
| Signing private key path              | ./.././././keys/oem_private_key_0.pem                                 |
| Hardware Security & Compliance        |                                                                       |
| HW security counter number            | 3                                                                     |
| HW security counter value             | 1                                                                     |
| Project 2 Configuration               |                                                                       |
| Memory Slots                          |                                                                       |
| RAM Load                              |                                                                       |
| Build & Deployment                    |                                                                       |
| Project name                          | proj_cm55                                                             |
| Input hex file path                   | ./.././././build/project_hex/proj_cm55.hex                            |
| Output hex file path                  | ./.././././build/project_hex/proj_cm55_signed.hex                     |
| Signing private key path              | ./.././././keys/oem_private_key_0.pem                                 |
| Hardware Security & Compliance        |                                                                       |
| HW security counter number            | 4                                                                     |
| HW security counter value             | 1                                                                     |

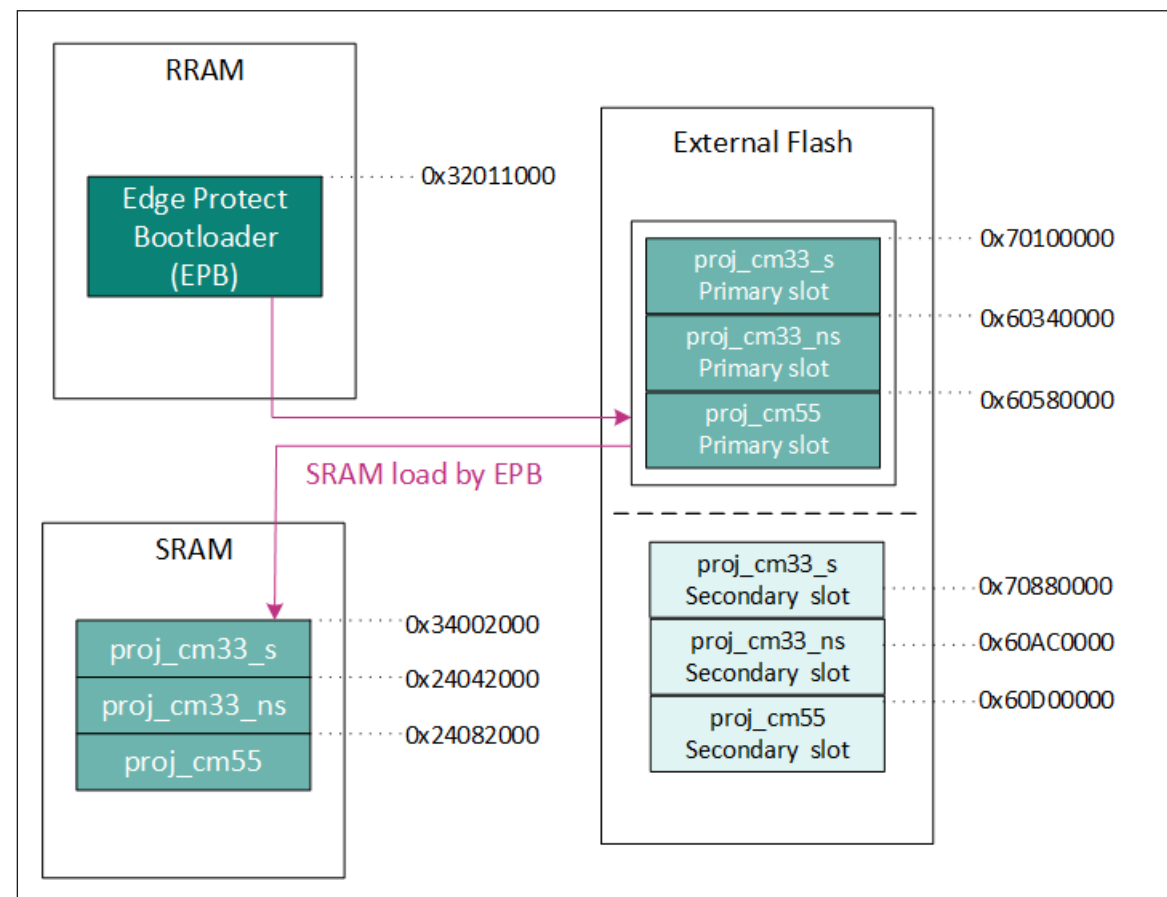
# Dependency check

- When enabled, EPB performs dependency checks and rejects incompatible updates
- Useful feature when only one of the application is being updated
- Set image dependencies in the format: (image\_ID, image\_version) where the image\_ID on which this image depends. It starts from 0. image\_version version on which this image depends. One image can depend on the several other images such as (1, 1.1.0), (2, 2.0.0)



## SRAM loading by EPB

- Execution of user application from the external flash has some drawbacks such as lower execution speed, higher power consumption etc., when compared to execution from SRAM
- Bootloader supports the RAM loading of application for execution from SRAM memory. In this case, the application images are programmed to persistent external flash memory, but copied into the internal RAM before execution
- Application should be compiled for execution from SRAM memory
- Refer to [PSOC™ Edge SRAM Loading](#) code example





# Encryption support

- Edge Protect Bootloader supports the encrypted firmware when EPB runs from RRAM only.
- Encrypted user images must be located in the external flash memory and executed in SMIF XIP mode or loaded to SRAM for execution.
- Edge Protect Bootloader supports the execution of the encrypted application in the following different methods
  - Single key encryption (EPC 2)
  - Multi key encryption (EPC 2)
  - Secure encryption (EPC 4)
- For more details check out the Encryption training module.

## Measured boot and data sharing

- Mechanism for sharing boot status information (also known as measured boot) and an interface for sharing application specific information with the runtime software in a shared memory region
- When enabled, adds a BOOT\_RECORD TLV to the image manifest. The TLV consists of the following information about the image in CBOR encoded format
  - Software type (role of the software component)
  - Software version
  - Signer ID (identifies the signing authority)
  - Measurement value (hash of the image)
  - Measurement type (algorithm used to calculate the measurement value)
- Measured boot and data sharing feature by the bootloader is used by other components such as Trusted Firmware-M for attestation services
- Refer to TF-M attestation code examples for [EPC 2](#) and [EPC 4](#) demonstration of how to use this feature along with TF-M

# Serial logging

- The bootloader logs the status and reports over the serial port during the boot and upgrade process.
- The logs are printed over an UART port and are determined based on the minimum logging level that is set during the compile time.
- Five logging levels are supported :  
 MCUBOOT\_LOG\_LEVEL\_OFF (0)  
 MCUBOOT\_LOG\_LEVEL\_ERROR (1)  
 MCUBOOT\_LOG\_LEVEL\_WARNING (2)  
 MCUBOOT\_LOG\_LEVEL\_INFO (3)  
 MCUBOOT\_LOG\_LEVEL\_DEBUG (4)

| Edge Protect Bootloader Solution - Parameters |                                        |
|-----------------------------------------------|----------------------------------------|
| Enter filter text...                          |                                        |
| Name                                          | Value                                  |
| ▼ Documentation                               |                                        |
| ① Edge Protect Bootloader documentation       | <a href="#">EdgeProtect Bootloader</a> |
| ▼ MCUBoot config                              |                                        |
| ① Upgrade mode                                | overwrite                              |
| ① Logging level                               | dbg                                    |
| ① FIH level                                   | off                                    |
| ① Image count                                 | 3                                      |
| ① Validate boot slot                          | <input type="checkbox"/>               |
| ① Validate upgrade slot                       | <input type="checkbox"/>               |
| ① Measured boot                               | <input type="checkbox"/>               |
| ① HW rollback protection                      | <input type="checkbox"/>               |
| ① Dependency check                            | <input type="checkbox"/>               |
| ① SW downgrade prevention                     | <input type="checkbox"/>               |
| ① Stage SE RAM App                            | <input type="checkbox"/>               |
| ① Watchdog                                    | <input type="checkbox"/>               |

# Fault Injection Hardening (FIH)

- To mitigate hardware-based fault injection attacks (e.g., power glitching, EM pulses) aimed at executing unauthorized code, MCUboot incorporates software countermeasures
- Their activation is controlled by the `MCUBOOT_FIH_PROFILE_*` macros, offering four protection levels: none, moderate, medium, and maximum
- The fault injection mitigation library has support for different measures, which can either be enabled/disabled separately or by defining one of the `MCUBOOT_FIH_PROFILE`

| Edge Protect Bootloader Solution - Parameters |                                                               |
|-----------------------------------------------|---------------------------------------------------------------|
| Enter filter text...                          |                                                               |
| Name                                          | Value                                                         |
| ▼ Documentation                               |                                                               |
| ① Edge Protect Bootloader documentation       | <a href="#">EdgeProtect Bootloader Configuration Overview</a> |
| ▼ MCUBoot config                              |                                                               |
| ① Upgrade mode                                | overwrite                                                     |
| ① Logging level                               | dbg                                                           |
| ① FIH level                                   | off                                                           |
| ① Image count                                 | 3                                                             |
| ① Validate boot slot                          | <input type="checkbox"/>                                      |
| ① Validate upgrade slot                       | <input type="checkbox"/>                                      |
| ① Measured boot                               | <input type="checkbox"/>                                      |
| ① HW rollback protection                      | <input type="checkbox"/>                                      |
| ① Dependency check                            | <input type="checkbox"/>                                      |
| ① SW downgrade prevention                     | <input type="checkbox"/>                                      |
| ① Stage SE RAM App                            | <input type="checkbox"/>                                      |
| ① Watchdog                                    | <input type="checkbox"/>                                      |

## SE RT services update

- In EPC 4 devices, the Secure Enclave runtime services (SE RT) library may be updated using the SE RAM App staging feature of EPB
- This feature allows users to use the EPB to stage the SE RT Services update RAM application from a persistent storage region and indicate to the ROM Boot to execute it
- The SE RAM App must be placed in a storage region (using OTA or other mechanism)

| Edge Protect Bootloader Solution - Parameters |                                                               |
|-----------------------------------------------|---------------------------------------------------------------|
| Enter filter text...                          |                                                               |
| Name                                          | Value                                                         |
| ▼ Documentation                               |                                                               |
| Edge Protect Bootloader documentation         | <a href="#">EdgeProtect Bootloader Configuration Overview</a> |
| ▼ MCUBoot config                              |                                                               |
| Upgrade mode                                  | overwrite                                                     |
| Logging level                                 | dbg                                                           |
| FIH level                                     | off                                                           |
| Image count                                   | 3                                                             |
| Validate boot slot                            | <input type="checkbox"/>                                      |
| Validate upgrade slot                         | <input type="checkbox"/>                                      |
| Measured boot                                 | <input type="checkbox"/>                                      |
| HW rollback protection                        | <input type="checkbox"/>                                      |
| Dependency check                              | <input type="checkbox"/>                                      |
| SW downgrade prevention                       | <input type="checkbox"/>                                      |
| Stage SE RAM App                              | <input checked="" type="checkbox"/>                           |
| ▼ SE RAM App Staging                          |                                                               |
| SE RAM App staging storage region             | SE_RAM_App (Memory Region 28)                                 |
| SE RAM App staging RAM address                | 0x34002000                                                    |

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# Running Edge Protect Bootloader on RRAM

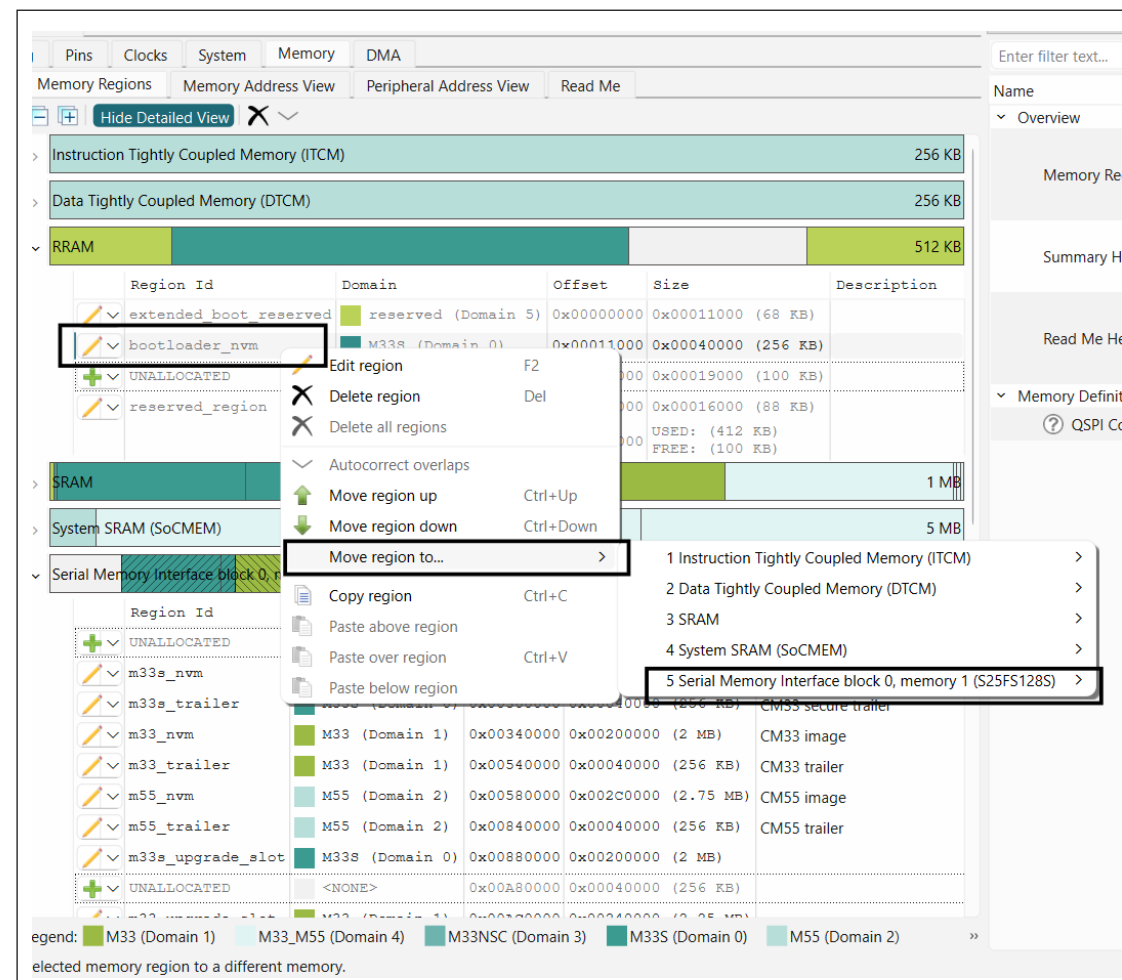
- Edge Protect Bootloader running on RRAM
  - Edge Protect Bootloader programmed to RRAM
  - Launched by extended boot on RRAM
  - Execute in-place (XIP) from RRAM
- This is the default configuration of [Edge Protect Bootloader](#)
- **Advantages**
  - Better resilience against attacks
  - Low-power
  - Faster boot to first OEM owned code
- **Disadvantages**
  - Limited size of RRAM

- Extended boot can launch the Edge Protect Bootloader from RRAM if BOOT SW (GPIO P17.6) EVK is set to LOW/OFF and the policy supports booting from external flash memory

*Note: "bootloader\_nvm" region must start at 0x00011000 offset to be successfully launched by the Extended Boot. If you want to place your first application in offsets other than 0x00011000, then you must provision the device with updated OEM policy file*

# Running Edge Protect Bootloader on external flash

- Edge Protect Bootloader running on external flash
  - Edge Protect Bootloader programmed to external flash
  - Launched by Extended boot on external flash
  - Execute in-place (XIP) from external flash
- BOOT SW pin (GPIO P17.6) on the EVK must be in the ON/HIGH position
- *Note: "bootloader\_nvm" region must start at 0x00100000 offset to be successfully launched by the Extended Boot. If you want to place your first application in offsets other than 0x00100000, then you must provision the device with updated OEM policy file*





# Running Edge Protect Bootloader on SRAM

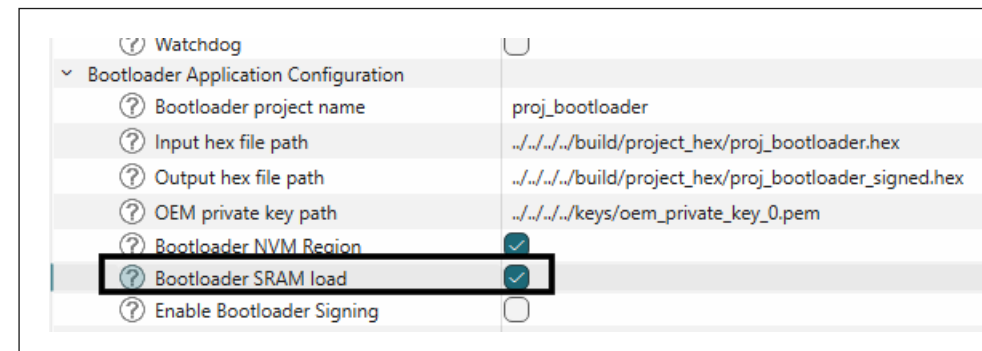
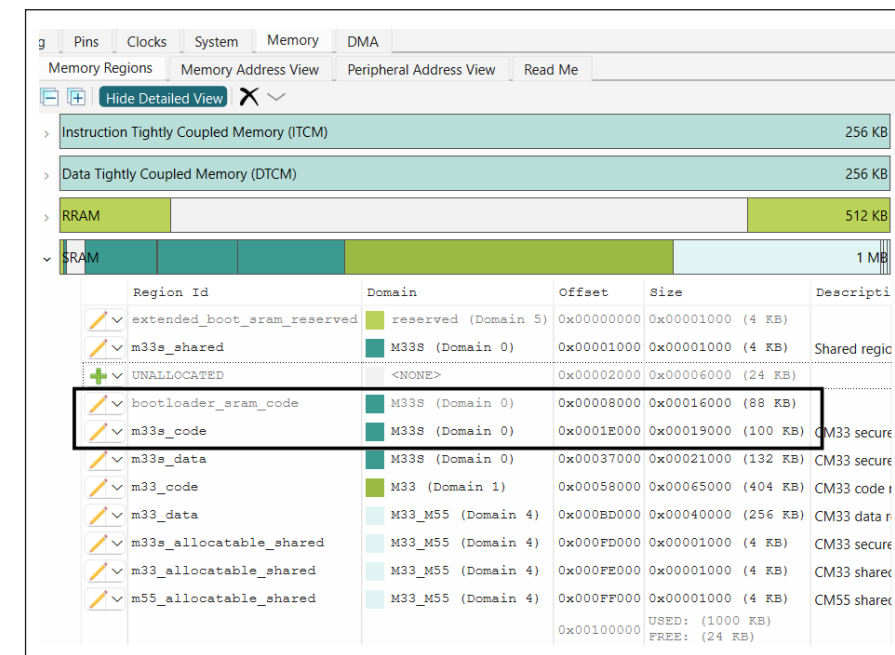
- Edge Protect Bootloader running on SRAM
  - Edge Protect Bootloader programmed to external flash
  - Copied to SRAM (and authenticated in SRAM, if Secure boot is enabled) by extended boot
  - Execute from SRAM

## Advantages

- Reduced security risks compared to running code from external flash
- Saves ~160 KB of internal RRAM for user application, without compromising on security

## Disadvantages

- Consumes a large chunk of SRAM region for code and data
- Application may reuse the SRAM region, provided that a secure application (such as TF-M) applies the modified security settings associated with the allocated region

The screenshot shows the 'Memory Regions' view in the IDE. The 'SRAM' region is expanded, showing various sub-regions. The 'bootloader\_sram\_code' and 'm33s\_code' regions are highlighted with a black box.

| Region Id                   | Domain              | Offset     | Size                | Description   |
|-----------------------------|---------------------|------------|---------------------|---------------|
| extended_boot_sram_reserved | reserved (Domain 5) | 0x00000000 | 0x00001000 (4 KB)   |               |
| m33s_shared                 | M33S (Domain 0)     | 0x00001000 | 0x00001000 (4 KB)   | Shared region |
| UNALLOCATED                 | <NONE>              | 0x00002000 | 0x00006000 (24 KB)  |               |
| bootloader_sram_code        | M33S (Domain 0)     | 0x00008000 | 0x00016000 (88 KB)  |               |
| m33s_code                   | M33S (Domain 0)     | 0x0001E000 | 0x00019000 (100 KB) | CM33 secure   |
| m33s_data                   | M33S (Domain 0)     | 0x00037000 | 0x00021000 (132 KB) | CM33 secure   |
| m33_code                    | M33 (Domain 1)      | 0x00058000 | 0x00065000 (404 KB) | CM33 code r   |
| m33_data                    | M33_M55 (Domain 4)  | 0x000BD000 | 0x00040000 (256 KB) | CM33 data r   |
| m33s_allocatable_shared     | M33_M55 (Domain 4)  | 0x000FD000 | 0x00001000 (4 KB)   | CM33 secure   |
| m33_allocatable_shared      | M33_M55 (Domain 4)  | 0x000FE000 | 0x00001000 (4 KB)   | CM33 sharec   |
| m55_allocatable_shared      | M33_M55 (Domain 4)  | 0x000FF000 | 0x00001000 (4 KB)   | CM55 sharec   |

USED: (1000 KB)  
FREE: (24 KB)

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# LAB: Secure Boot with Edge Protect Bootloader

- Set up the Basic Secure App to function alongside the Edge Protect Bootloader
- Enable secure boot in Edge Protect Bootloader

# Revision history

| Document revision | Date       | Description of change |
|-------------------|------------|-----------------------|
| **                | 2025-12-11 | Initial release.      |

